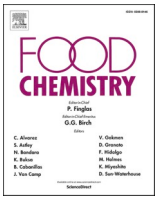
Food Chemistry 405 (2023) 134817



Contents lists available at ScienceDirect

Food Chemistry

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Extraction of weak hydrophobic sulforaphane from broccoli by salting-out assisted hydrophobic deep eutectic solvent extraction

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ARTICLE INFO

Keywords:
Sulforaphane
Broccoli
Deep eutectic solvent
Hydrophobic
Kamlet-Taft parameters
Density functional theory

Chemical compounds studied in this article:
Sulforaphane (PubChem CID 5350)
Methyl trioctyl ammonium chloride (PubChem CID 21218)
Ethylene glycol (PubChem CID 174)
Glycerol (PubChem CID 753)
1,4- Butanediol (PubChem CID 8064)
N-octanoic acid (PubChem CID 379)
N-capric acid (PubChem CID 2969)
Thymol (PubChem CID 6989)
Lidocaine (PubChem CID 3676)

ABSTRACT

In order to extract sulforaphane (SFN) from broccoli via green and efficient ways, a novel method based on salting-out assisted deep eutectic solvent (DES) has been developed. Compared to known organic solvent- (such as dichloromethane, ethyl acetate, *n*-hexane, etc.) based liquid–liquid extraction, this new $N_{8881}Cl$-based DES method exhibited excellent extraction efficiency for SFN, including a significant improvement due to the salting-out effect of KH_2PO_4. Under optimal conditions, 97.77 % of SFN was extracted by $N_{8881}Cl$-EG DES and more than 82.5 % of SFN was recovered by activated carbon from DES. In addition, further studies with Kamlet-Taft parameters and density functional theory showed that the H-bond accepting capacity of hydrophobic DES, the existing vdW interaction, and the electrostatic interaction between $N_{8881}Cl$-EG DES all contributed to efficient extraction of SFN. This is the first time that the underlying mechanism for SFN extraction by DES was revealed.

1. Introduction

Sulforaphane (SFN, 1-isothiocyanate-4-methanesulfonylbutane), an isothiocyanate derived from glucoraphanin (GR), has attracted considerable attention due to its antioxidant, anti-aging, anti-inflammatory, and anticarcinogenic properties. (Calabrese & Kozumbo, 2021). SFN has been suggested as one of the best natural inducers of human phase II enzymes that can detoxify cancer-causing chemicals thus protecting against carcinogens and inflammation (Kamal, Akter, Lin, & Nazzal, 2020). Currently, SFN can be obtained via total synthesis, semi-synthesis, or extraction from natural plants (Vo, Truong, Tran, Do, Pham, & Thai, 2016). Due to its wide availability, natural plant-derived SFN has been emerged as a very promising functional ingredient in functional food products and beverages, cosmeceuticals, and medicines and nearly 100 products containing SFN can be found at online stores (https://www.amazon.com), such as BroccoMax (Jarrow Formulas), SULFORAPHAN (Carlyle), Crucera-SGS (Thorne Research), etc. Thus, there is considerable interest in extracting SFN from natural plants.

Broccoli is an important crop worldwide with a large amount of production and consumption annually (approximately 26.92 million

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https://doi.org/10.1016/j.foodchem.2022.134817
Received 12 June 2022; Received in revised form 25 October 2022; Accepted 29 October 2022
Available online 4 November 2022
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tons in 2019) (Li, Xia, Liu, Guo, He, Liu, et al., 2021). Broccoli sprouts have higher SFN than mature broccoli, and about 0.48 to 2.3 mg/g of SFN were detected from dry broccoli sprouts according to literature (Cai, Augustin, Jegasothy, Wang, & Terefe, 2020; Liu, Wu, Xu, Wang, Wang, Zhang, et al., 2022). However, SFN is mainly found in the form of its glucosinolate precursor (GR), which is converted to SFN via hydrolysis of GR by myrosinase enzyme (thioglucoside glucohydrolase, EC 3.2.3.1) after plant tissues or cells were damaged (Van Eylen, Oey, Hendrickx, & Van Loey, 2007). Traditionally, the preparation process of SFN from broccoli sprouts included hydrolysis by aqueous extraction under heating, and then purification by liquid–liquid extraction with organic solvents such as hexane, ethyl acetate, dichloromethane, methyl *t*-butyl, etc. However, these organic solvents lack-satisfactory extraction efficiency due to the weak hydrophobicity of SFN (log P = 0.23) (Ares, Bernal, Martín, Bernal, & Nozal, 2014; Tříska, Balík, Houška, Novotná, Magner, Vrchotová, et al., 2021). Although multiple extraction methods have been reported to improve the SFN yield, including high hydrostatic pressure and ultrasound extraction method (Briones-Labarca, Plaza-Morales, Giovagnoli-Vicuña, & Jamett, 2015), high-pressure homogenization extraction method (Xing, Cheng, Chen, Shan, Ruan, Li, et al., 2019), and microwave-assisted extraction method (Tanongkankit, Sablani, Chiewchan, & Devahastin, 2013). These methods inevitably adopted organic solvents, which were not green. Therefore, it remains as a challenge to find an efficient and green extraction method to obtain high-yield SFN from broccoli.

As a new generation of green solvent, deep eutectic solvents (DESs) are the complexation of hydrogen bond donor (HBD) and hydrogen bond acceptor (HBA) compounds, which exist as a eutectic mixture with a melting point lower than each constituent. DESs exhibit many advantages, including low price, high safety, and tunable physical and chemical properties, so they have been widely used to extract valuable active ingredients, such as polyphenols (Peng, Duan, Yao, Zhang, Zhao, Zu, et al., 2016) and anthraquinones (Deng, Zong, & Xiao, 2017) from natural plants.

However, most reported DESs are hydrophilic with strong polarity and high viscosity, making it difficult for efficient extraction of SFN from broccoli hydrolysate. In 2015, van Osch et al. (van Osch, Zubeir, van den Bruinhorst, Rocha, & Kroon, 2015) first reported the first hydrophobic DES for the extraction of weakly polar fatty acids from aqueous solution, which was based on long-chain quaternary ammonium salts. Subsequently, a series of hydrophobic DESs were developed for extraction of hazardous substances from environmental samples (Fattahi, Pirsaheb, Moradi, Mohebbi, Karimi, & Hashemi, 2022; Fattahi, Shamsipur, Nematifar, Babajani, Moradi, Soltani, et al., 2022; Habibollahi, Karimyan, Arfaeinia, Mirzaei, Safari, Akramipour, et al., 2019), and personal care products (Fattahi, Hashemi, Shiri, Shamsipur, & Babajani, 2022). Recently, several hydrophobic DES have been developed for extraction of active ingredients (indoles, artemisinin, polyprenyl acetates) from plants (Cao, Yang, Cao, Wang, & Su, 2017a, 2017b; Koel, Kuhtinskaja, & Vaher, 2020). These previous studies indicated that natural hydrophobic DESs might serve as a green alternative for extraction of SFN from natural herbs.

Nevertheless, in comparison to reported weakly polar indoles (log P of 3.7) (Koel, Kuhtinskaja, & Vaher, 2020) and artemisinin (log P more than 2) (Shahzad, Shah, Ansari, Riaz, Safdar, Hussain, et al., 2012), SFN has a significantly weaker hydrophobicity with log P of 0.23 (Franklin, Dickinson, Karlage, Bowden, & Myrdal, 2014), which is a major challenge for the extraction of SFN by natural hydrophobic DESs from broccoli hydrolysate. In this study, a novel salting-out assisted DES extraction method was developed for extraction and purification SFN from broccoli for the first time. First, a series of natural hydrophobic DESs were prepared and characterized by FT-IR and ^{1}H NMR spectroscopies. Afterwards, these DESs were adopted as extraction solvents to isolate SFN from broccoli hydrolysate and their efficiencies were systematically compared with traditional organic solvents. Next, salting-out assisted hydrophobic DES extraction was constructed and the extraction conditions were optimized with One Variable-at-a-Time (OVAT) and response surface methodology (RSM). Under the derived optimal extraction conditions, the recovery of SFN from DESs were examined. In addition, Kamlet-Taft (K-T) parameters of DESs and calculations with density functional theory (DFT) were conducted to explore the underlying mechanism of SFN extraction. To the best of our knowledge, this is the first time that a salting-out assisted DES extraction method was successfully developed for extraction of weakly hydrophobic compounds, along with the mechanism uncovered by K-T parameters and DFT analysis of this phenomenon.

2. Materials and methods

2.1. Materials

Broccoli was obtained from a local agricultural market (Wunan Community, Wuhan City, Hubei Province, China). Sulforaphane standard (SFN, 95 %), Thymol (Thy, 99.0 %), Methyltrioctylammonium Chloride (N_{8881}Cl, 97 %), 1,4-Butanediol (But, 98 %), Nile Red (95 %) were purchased from Macklin biochemical technology Co., ltd. (Shanghai, China). DL-menthol (Men, 98 %), *n*-decanoic acid (DA, 99 %), *n*-hexanoic acid (HA, 99 %), *n*-octanoic acid (OA, 99 %) were purchased from Aladdin Biochemical Technology Co., ltd. (Shanghai, China). Glycerol (Gly, 99 %). Ethylene glycol (EG, 99 %), dichloromethane (DCM, 99 %), ethyl acetate (EAC, 99 %), *n*-hexane (NHA, 99 %) were supplied by Sinopharm Chemical Reagent Co. (Shanghai, China). Macroporous resin D101, DM-130, AB-8, ADS-17, HPD-17, HPD-450 and activated carbon (AC) were provided by Baoen Adsorption Material Technology Co., ltd. (Cangzhou, China). Deionized water was obtained from a Milli-Q system (Millipore, Molsheim, France). All other reagents used were of analytical grade.

2.2. Preparation and characterization of natural hydrophobic DESs

N_{8881}Cl, Men and DA were used as HBA, and HA, Thy, Oct, But, Gly and EG were employed as HBD. DES were synthesized according to literature (Deng, Zong, & Xiao, 2017; Sun, Zheng, Yang, Zhang, Cheng, Ghiladi, et al., 2021). First, HBA and HBD were mixed in a required molar ratio, and then the obtained mixture was heated in an 80 °C oil bath with magnetic stirring (500 rpm) to obtain a homogeneous and transparent liquid DES at room temperature. Viscosities of DES were measured by using a Rheometer (DISCOVERY HR-2, USA) at 25 °C with a shear rate of 0.1–300 s-1. Densities of DESs were determined by aspirating 100 μL of DES with a pipette at 25 °C, and then the added mass was weighed by an analytical balance (Shimadzu, Japan). This whole procedure was repeated six times and the density was calculated according to $\rho = \Delta m/V$. Polarities of DESs were determined based on the pyrene fluorescence probe method (Huang, Deng, Li, Zheng, Wang, & Xiao, 2022) and the parameter (I1/I3) was calculated to evaluate the polarity. FT-IR spectra were collected on a FT-IR spectrometer (Thermo Scientific Nicolet iS10, USA). 1H NMR spectra were recorded at 9.40 T of magnetic field intensity, 90o of pulse angle at R.T. on a 400 MHz Bruker DPX-400 (Germany) with a sealed capillary tube (containing deuterated water) as internal standard.

2.3. Preparation of broccoli hydrolysate and extraction of SFN

Preparation of broccoli powder Broccoli florets were cut from broccoli and washed for 1 min to remove surface contaminants, and stored frozen at −80 °C for 12 h. The frozen florets were freeze-dried at 20 °C under a vacuum of 200 Pa for 12 h using a FreeZone12L Freeze Dryer (Labconco, USA). Then they were grounded and passed through a 60-mesh sieve to obtain dried broccoli florets powder.

Preparation of broccoli hydrolysate To convert GR into SFN, broccoli hydrolysate sample was prepared according to previously reported protocols (Wei, Liu, Zheng, & Zheng, 2020) with slight modifications.

0.1 g of dried broccoli florets powder was dispersed in 2 mL of deionized water and stirred (500 rpm) at 45 °C for 1 h. Then the mixture was centrifuged at 10,000 rpm for 5 min, and the supernatant (broccoli hydrolysate) was collected for subsequent experiments.

Extraction of SFN by salting-out assisted DES extraction KH_2PO_4 (0.125 g) was dissolved in 500 μL of broccoli hydrolysate to a final salt concentration of 0.25 g/mL), and then 500 μL DES was added. After vortexing for 30 s, the mixture was centrifuged at 14,000 rpm for 5 min and a liquid–liquid two phase system appeared with hydrophobic DESs enriched in the top phase. The volumes of each liquid phase were recorded and the top phase was collected for HPLC analysis. The extraction efficiency (EE) of SFN was calculated by the following equations:

$$EE(\%) = \frac{C_D \times V_D}{C_O \times V_O} \times 100\% \quad (1)$$

where V_D and V_O are the volume of the DES-rich phase and broccoli hydrolysate, respectively; C_D and C_O represent the concentration of SFN in the DES-rich phase and broccoli hydrolysate, respectively.

2.4. Recovery of SFN from DES extract

The recovery of SFN from DES extract was achieved by employing a slightly modified macroporous resin method (Cao, Yang, Cao, Wang, & Su, 2017b). First, macroporous resins and activated carbon were soaked with ethanol, NaOH solution and HCl solution in sequence (the detailed method is shown in Appendix). Then, 0.1 g of resins was mixed with 1.0 mL of hydrophobic DES extract. After shaking at 150 rpm (R.T.) for 12 h, the mixture was centrifuged at 10,000 rpm for 5 min. The supernatant was collected and the resin-bound SFN were filtered out and eluted with 1.0 mL of solvent at 150 rpm for 12 h. The content of SFN in DES extract, DES extract after adsorption, and the eluate were determined by HPLC.

2.5. HPLC analysis of SFN

Chromatographic analysis was carried out on a LC-20A HPLC system (Shimadzu, Japan) equipped with a UV detector and a quaternary pump. Chromatographic separation of SFN was performed on an Amethyst C_{18} column (250 mm × 4.6 mm, 5 μm) at 30 °C. The mobile phase was composed of water (eluent A) and acetonitrile (eluent B) with a gradient elution. To elute the impurities quickly and recover baseline and column pressure, a gradient program was set as follows: 0–13 min at 20 % B, 13–17 min 20 %-100 % B, 17–20 min 100 % B, 20–25 min 20 % B. The flow rate was set at 1.0 mL/min, the detection was at 254 nm and the injected volume was 10 μL. A calibration curve ($y = 3140.8x + 3652.9$, $n = 7$) was obtained by fitting the concentration of SFN standard to its peak area. The intra-day and inter-day precision evaluation of the recovery rate was performed with the standard solutions of SFN at different concentrations. The validation results of quantification method was shown in Table S1.

2.6. Measurement of the K-T parameters

K-T parameters of DESs were measured according to literature with Nile Red, 4-nitroaniline (NH_2) and N,*N*-diethyl-4-nitroaniline (NEt_2) as a solvatochromic indicator to determine α, β and π^*, respectively (Xia, Liu, Meng, Cheng, Chen, Liu, et al., 2018). Briefly, 100 μL of solvatochromic indicator solution (50 μM in ethanol) was added to a 1.5 mL sample vial, and ethanol was evaporated naturally at room temperature. Then, 1 mL of DESs or organic solvent was added and the solution was sonicated for 10 min. The absorption spectrum of solution was detected using a Carry60 UV spectrophotometer (Agilent, USA) in the range of 400–700 nm for Nile Red and 300–500 nm for NH_2 and NEt_2. The parameters of α, β and π^* were calculated according to the following equations:

$$\pi^* = 0.314 \times (27.52 - \nu_{NEt2}) \quad (2)$$

$$\beta = 11.134 - \frac{3580}{\lambda(NH_2)_{max}} - 1.125\Pi^* \quad (3)$$

$$\alpha = \frac{(19.9657 - 1.0241\Pi^* - \nu_{NR})}{1.6078} \quad (4)$$

Where λ^{max} is the maximum UV absorption of solvatochromic indicator and $\nu = 1/(\lambda^{max} \times 10^{-4})$.

2.7. Simulation

Molecular simulations and DFT were adopted to explore the molecular interaction between DESs and extracts according to previous studies (Gutiérrez, Aparicio, & Atilhan, 2019; Makoś, Słupek, & Gębicki, 2020). First, the initial molecular structures were built by Gaussian View 5.0. Then, structure optimization, the HOMO-LUMO energy, and interaction energy calculation between DESs and SFN were conducted using M06-2X/6-31G* of the ORCA 5.0.3 software package. The counterpoise procedure was used to account for basis set superposition error (BSSE). For the effective visualization of the interaction between DESs and SFN, the electrostatic potential field (ESP) was adopted to explore the electronic property of stationary points, and the reduced density gradient (RDG) was used to discuss the nature and strength of interactions displayed in the isosurface of DESs and SFN using the Multiwfn program code. The Visual Molecular Dynamics 1.9.3. software was obtained to graphically present the results.

3. Results and discussion

3.1. Characterization of DESs

The composition and physicochemical properties of eight natural hydrophobic DESs are summarized in Table 1. Densities of these DESs were between 0.860 and 0.946 g•cm^{-3}, less than that of water, indicating that these DESs will be enriched in the top phase after mixing with water. Viscosities of long-chain alkyl acid-based nonionic hydrophobic DESs (Men-HA, Men-OA, DA-Thy and DA-Oct) were in the range of 6.43–10.09 mPa·s, which were consistent with reported nonionic hydrophobic DESs (4.03–22.42 mPa·s) (Huang, Deng, Li, Zheng, Wang, & Xiao, 2022). Long-chain quaternary (N8881Cl) based hydrophobic DESs displayed higher viscosity (124.82–408.91 mPa·s) than nonionic hydrophobic DESs, which was due to the presence of coulombic forces in DES (Huang, Deng, Li, Zheng, Wang, & Xiao, 2022). The polarity of hydrophobic DESs (I1/I3) ranged from 0.82 to 1.09, which were significantly lower than the polarity of methanol (1.437) and higher than that of *n*-hexane (0.675), indicating that the prepared natural hydrophobic DESs show moderate polarity and might have a strong affinity for moderately polar SFN (log P of 0.23) (Franklin, Dickinson, Karlage, Bowden, & Myrdal, 2014). In addition, the hydrophobicity of these nonionic DESs and N8881Cl based DES has been well studied in literature (Huang, Deng, Li, Zheng, Wang, & Xiao, 2022; Tang, Liu, Yin, Zhu, Huang, Dong, et al., 2021).

FT-IR and ^{1}H NMR spectroscopies were adopted to verify the intermolecular interaction in DESs. The results (Fig. S1-8) showed that all peaks of natural hydrophobic DESs were attributed to their initial components and no new peaks of covalent bonds appeared, indicating the absence of any chemical reaction during the preparation of DESs. The FT-IR spectra of DESs (Fig. S1a-S8a) revealed that the O—H stretching peak in DESs broadened and shifted to a lower wavenumber compared to that of glycol, menthol, butanediol, glycerol, thymol and alkanols band (e.g., the hydroxyl band of ethylene glycol shifted from 3387.35 cm^{-1} to 3367.19 cm^{-1} in N_{8881}Cl -EG DES, Fig. S1a), which demonstrated the transfer of the electron cloud of an oxygen atom to hydrogen bonding and confirmed the existence of hydrogen bonds in

Table 1
Compositions and physicochemical properties of the studied DES.

No.	HBA	HBD	Molar Ratio	Viscosity (mPa·s)	Density (g mL^{-1})	I_1/I_3	Abb.
1	Menthol	Hexanoic Acid	1:1	10.09	0.90	0.82	Men-HA
2	Menthol	Octanoic Acid	1:1	11.87	0.90	0.83	Men-OA
3	Decanoic Acid	Thymol	1:1	12.22	0.93	0.87	DA-Thy
4	Decanoic Acid	Octanol	1:2	6.428	0.85	0.83	DA-Oct
5	N_{8881}Cl	Octanoic Acid	1:2	408.91	0.88	0.97	N_{8881}Cl -OA
6	N_{8881}Cl	Glycerol	1:2	350.21	0.98	1.06	N_{8881}Cl -Gly
7	N_{8881}Cl	Butanediol	1:2	244.82	0.91	1.09	N_{8881}Cl -But
8	N_{8881}Cl	Ethylene Glycol	1:2	124.82	0.92	1.06	N_{8881}Cl -EG

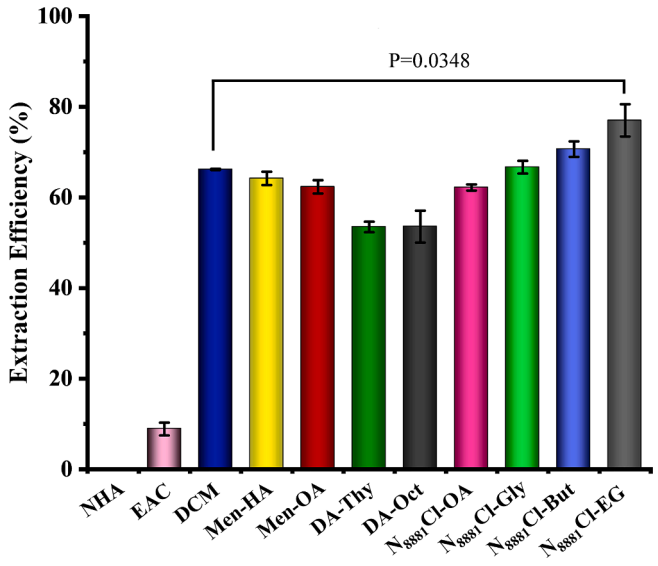


Fig. 1. Effect of different solvents on the extraction efficiency of SFN. NHA: *n*-hexane; EAC: ethyl acetate; DCM: dichloromethane; Men: DL-menthol; HA: *n*-hexanoic acid; OA: *n*-octanoic acid; DA: decanoic acid; Thy: thymol; Oct: Octanol; N_{8881}-Cl: ethyltrioctylammonium Chloride; Gly: glycerol; EG: ethylene glycol; But: 1,4-butanediol.

DES (Deng, Zong, & Xiao, 2017). In addition, the chemical shifts of H in DESs (Table S2-S9) showed that ^{1}H of –OH or –COOH (or close to –OH and –COOH) in DESs were shifted to high field (e.g., the H_6 of EG shifted from 5.30 ppm to 4.36 ppm in N_{8881}Cl-EG DES, Table S2). This is interpreted as the presence of hydrogen bonds to the –OH (or –COOH) groups results in a shielding effect (Fan, Wu, Cai, Yang, & Yang, 2020).

3.2. Comparison of extractability of DESs with different solvents

Extraction solvent impacts the partition coefficient of compounds in the liquid–liquid two-phase system, which affects efficient extraction. Here, eight types of natural hydrophobic DESs were evaluated for their ability to extract SFN from broccoli hydrolysate, in comparison to traditional organic solvents (DCM, NHA and EAC). As shown in Fig. 1, SFN could not be extracted by *n*-hexane, which was due to its poor solubility for SFN (Liang, Li, Yuan, & Vriesekoop, 2008). The extraction efficiency (EE) of all the eight DESs were more than 53.5 %, which were significantly better than that of EAC (8.9 %) and *n*-hexane. In particular, the EE of N8881Cl-EG was significantly higher than that of DCM ($p < 0.05$). The above results indicated that long chain alkanoic acid or quaternary ammonium salts based hydrophobic DESs were capable of efficiently extracting SFN from broccoli hydrolysate. Among the eight DESs, N_{8881}Cl based DESs showed higher EE for SFN, which allowed easier extraction of the weakly hydrophobic SFN. Considering that N_{8881}Cl-EG DES has the highest EE, it was adopted as extraction medium for the next-step of the investigation.

3.3. Optimizing extraction conditions of SFN

To improve the extraction efficiency, salting-out assisted extraction was adopted and the extraction conditions were optimized by OVAT experiment. The optimized factors included the molar ratio of HBA to HBD of DES (HBA/HBD, 2:1–1:2, mol/mol), type of salt ($(NH_4)_2SO_4$, Na_2SO_4, K_2HPO_4, KH_2PO_4, $Na_3C_6H_5O_7$), concentration of salt (0.15–0.35 g/mL), volume ratio of DES to broccoli hydrolysate (V_{DES}/V_{BH}, 1:2–2:1) and vortexing time (15–120 s).

3.3.1. Molar ratio of HBA to HBD of DES

In this work, N_{8881}Cl-EG DES with different HBA/HBD were tested to evaluate their ability to extract SFN from broccoli hydrolysate without the use of salts. The results (Fig. 2a) showed that the EE slightly increased initially and then decreased slightly with elevated HBA/HBD values. This phenomenon was also reported for extraction of polyprenyl acetates employing N_{8881}Cl based hydrophobic DES (Cao, Yang, Cao, Wang, & Su, 2017a). The increase in EE was attributed to the strengthened hydrophobic interaction between SFN and DES with the enhanced HBA content. When HBA/HBD exceeded 1:1, the viscosity of DES significantly increased (Fig. S9), which in turn hindered the mass transfer of SFN to DES enriched phase and resulted in the decreased EE of SFN. Considering the high EE for SFN, 1:1 of HBA/HBD was selected for the next-step investigation.

3.3.2. Type of salts solution

To explore the influence of salt types on the EE of SFN by salting-out assisted DES extraction (N_{8881}Cl-EG DES, HBA/HBD value of 1:1), different types of salt were introduced into broccoli hydrolysate to a final concentration of 0.25 g/mL. The results showed that, compared with the control group (i.e., without salt), the EE was increased via addition of $(NH_4)_2SO_4$, Na_2SO_4 and KH_2PO_4 (Fig. 2b). However, the EE was decreased upon introduction of $Na_3C_6H_5O_7$ or K_2HPO_4 due to the decomposition of SFN under alkaline conditions (Wu, Liang, Yuan, Wang, & Yan, 2010). Among these five salts, KH_2PO_4 provided the best phase separation ability and salting-out effect, thus the highest EE (93.6 %) (Zeng, Wang, Huang, Ding, Chen, & Xu, 2014). Therefore, KH2PO4 was selected for further investigations.

3.3.3. Concentration of salt

The effect of KH_2PO_4 concentration on the EE of SFN is shown in Fig. 2c. The EE increased rapidly at first, and then dropped slightly with increased KH_2PO_4 concentration. This phenomenon may be attributed to the enhanced salting-out effect with rising salt concentrations from 0.1 to 0.25 g/ml, leading to enrichment of SFN in the DES phase. Once salt concentrations were higher than 0.25 g/mL, viscosity started to inhibit the diffusion of SFN to the DES-rich phase, resulting in the slightly reduced EE (Deng, Zong, & Xiao, 2017). Thus, the optimal KH_2PO_4 concentration was set as 0.25 g/mL.

3.3.4. Volume ratio of DES to broccoli hydrolysate

The influence of different volume ratios of DES to broccoli hydrolysate (V_{DES}/V_{BH}, 1:2 to 2:1, mL/mL) on the EE of SFN was evaluated. The EE was improved significantly as V_{DES}/V_{BH} increased from 1:2 to

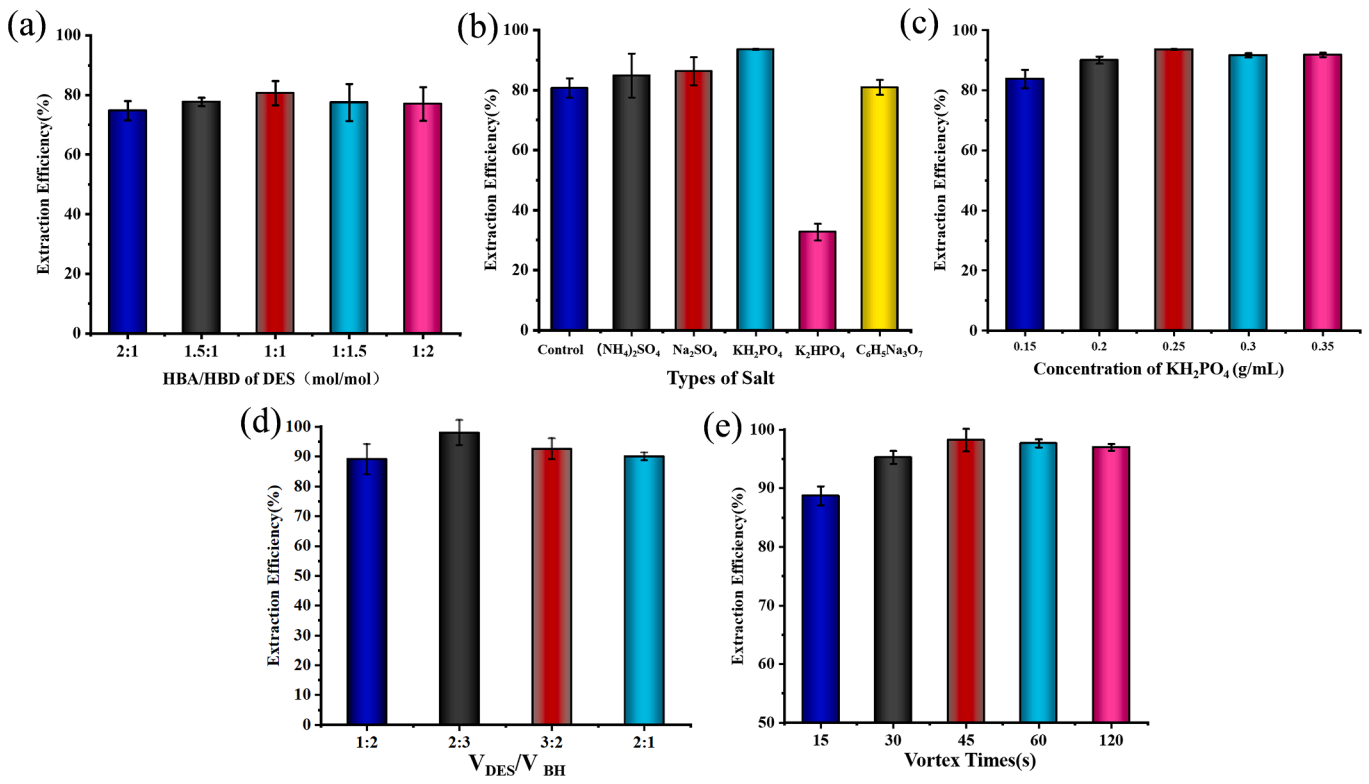


Fig. 2. Effect of HBA/HBD of DES (a), types of salt (b), K_2HPO_4 concentration (c), V_{DES}/V_{BH} (d) and vortex time (e) on the extraction efficiency of SFN.

2:3, and then decreased slightly with a further increase of V_{DES}/V_{BH} (Fig. 2d), indicating that SFN could be fully extracted at 2:3 of the V_{DES}/V_{BH}. In order to avoid wasting DES, a volume ratio of 2:3 V_{DES}/V_{BH} was selected.

3.3.5. Vortex time

Vortex performance depends on the rotational speed and duration. Here, the effect of vortex time (15–120 s) on the EE of SFN was investigated at a fixed rotational speed (nearly 2500 rpm). The results (Fig. 2e) revealed that the EE increased rapidly first and then reached a plateau at vortexing durations greater than 30 s. Thus, the optimal vortexing time was set at 30 s.

3.4. Screening of significant factors using Plackett-Burman design (PBD)

In order to evaluate the significance of factors on the extraction performance of salting-out assisted DES extraction of SFN, four experimental factors (HBA/HBD, KH_2PO_4 concentration, V_{DES}/V_{BH} and vortex time) were selected for PBD experiments. The detailed experimental domain of the variables are shown in Table S10. The obtained results for Plackett-Burman design (PBD) and the visualized Pareto chart are presented in Table S11 and Fig. S10, respectively. In Fig. S10, the bar length is proportional to its corresponding standardized affect value, and a factor was considered significant if it was greater than *p*-value (0.05), presented by the dotted line (Sun, et al., 2021). Among the four factors, HBA/HBD, KH_2PO_4 concentration and V_{DES}/V_{BH} were found to contribute most to the EE of SFN with *p*-values of 0.0307, 0.0364 and 0.0464, respectively (Table S12). Thus, these three factors were screened for subsequent RSM optimization and the vortex time (30 s) was kept in constant based on the OVAT results.

3.5. Optimization of extraction condition using RSM

Based on the results of OVAT optimization and PBD, factors and their ranges for RSM optimization by Box-Behnken experimental design (BBD) were set as follows: HBA/HBD of DES (A) from 3:2 to 1:2, KH_2PO_4 concentration (B) from 0.1 g/mL to 0.35 g/mL and V_{DES}/V_{BH} (C) from 1:2 to 2:1 (mL/mL). The EE of SFN was selected as response (R). The obtained experimental results of the BBD studies are shown in Table S13 with EE in the range of 78.41 % to 97.53 %. These data were then fitted to a second-order polynomial model, and the obtained regression model equation for the EE (R) and three variables were as follows:

$$R = 97.07 + 0.48A + 2.12B - 3.63C + 1.33AB - 2.79AC + 1.4BC - 4.46A^2 - 5.92B^2 - 6.38C^2$$

ANOVA was performed to evaluate the RSM model statistically. The results (Table S14) showed that the high F-value (638.34) and low *p*-value (<0.0001) indicated the model was significant and reliable enough to be applied for analysis of experimental data (Sun et al., 2021). The determination coefficient (R^2), adjusted determination coefficient (adjusted R^2) and predicted determination coefficient (predicted R^2) were 0.997, 0.979 and 0.882, respectively, demonstrating that the RSM model had a very high accuracy, reliability and excellent predictive capability. Among the interactional factors, AB, AC and BC were highly significant ($p < 0.022$), which might be attributed to the partition coefficient of SFN in hydrophobic DES liquid–liquid two phase depending on the properties of DES (affected by HBD/HBA), DES content (affected by V_{DES}/V_{BH}) and salting-out effect (affected by KH_2PO_4 concentration). In addition, the *p*-value (0.0586) of lack-of-fit was non-significant, indicating that the model was fit with credible prediction. Thus, the above results confirmed the RSM model was statistically ideal and its accuracy and reliability was adequate in this work.

In order to achieve the optimal extraction performance, the 3D response surface curves of RSM were plotted. As depicted in Fig. 3, the three response surface curves were all upper convex and contained the maximum point, indicating that the designed level ranges of factors were reasonable. Additionally, the changing trends of EE with factors were consistent with the results of the OVAT. Based on the RSM model, the optimum extraction conditions were set as follows: 1.08:1 of HBA/HBD of N_{8881}Cl-EG DES, 0.25 g/mL of KH_2PO_4 concentration and 1.04:1

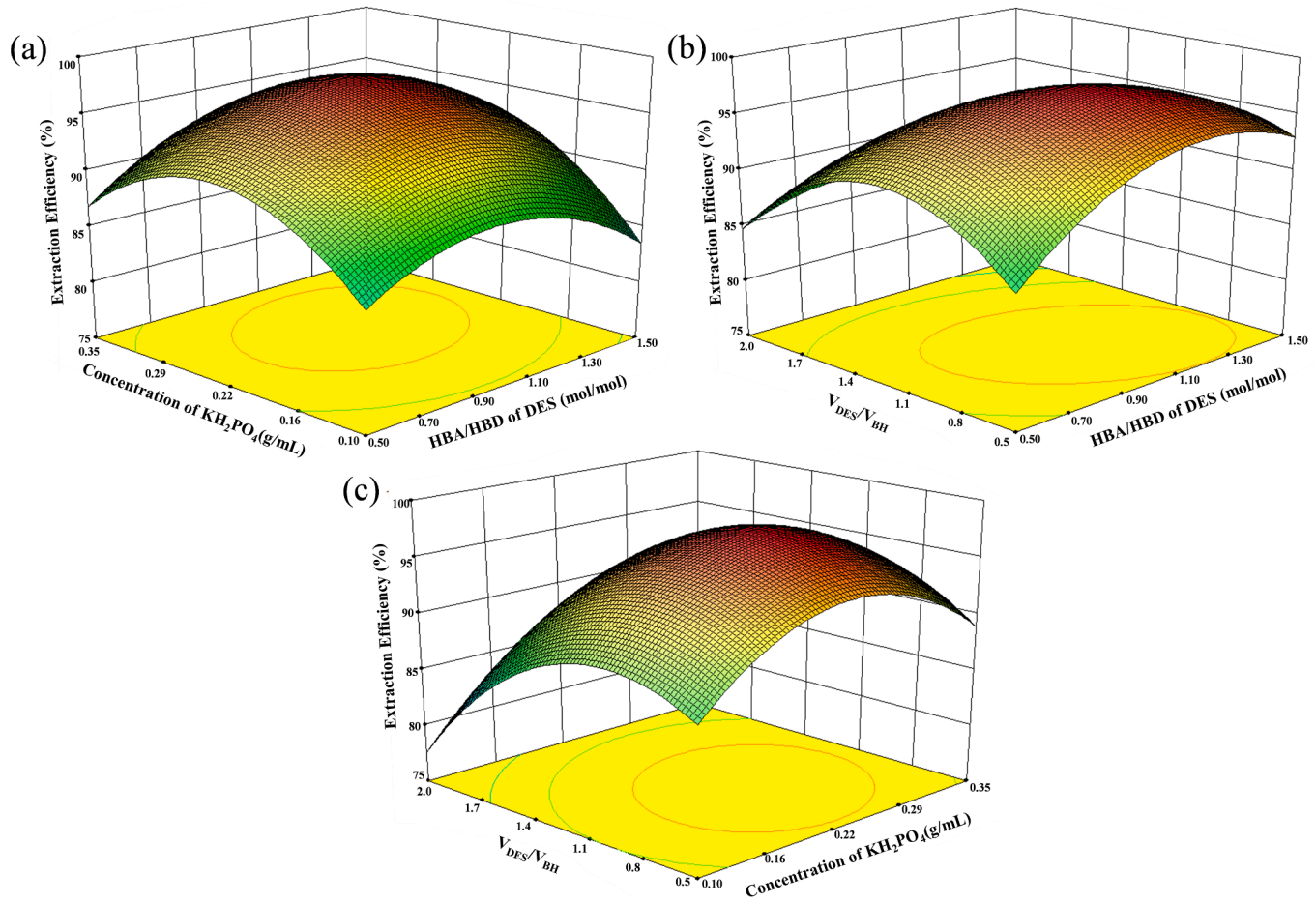


Fig. 3. Response surface plots for EE of SFN using BBD. (a): Mutual effects of KH_2PO_4 concentration and HBA/HBD of DES on EE of SFN; (b): Mutual effects of V_{DES}/V_{BH} and HBA/HBD of DES on EE of SFN; (c): Mutual effects of V_{DES}/V_{BH} and KH_2PO_4 concentration on EE of SFN.

of V_{DES}/V_{BH}, and the predicted value of EE was 97.83 %. For easy operation to verify the predicted EE, the extraction conditions were modified slightly (1:1 of HBA/HBD, 0.25 g/mL of KH_2PO_4 concentration and 1:1 of V_{DES}/V_{BH}). Under modified conditions, the experimental values (97.77 % ± 1.04, n = 3) were in good agreement with the predicted value (97.83 % ± 0.06 %), verifying the accuracy and reliability of the fitted model. In addition, to confirm the reliability of this method, a scale-up experiment was conducted under the optimized conditions. The results (Fig. S11) showed that the EE of SFN was more than 93.5 % even with different magnification (2, 5, or 10 folds), thus attesting to the reliability of our method.

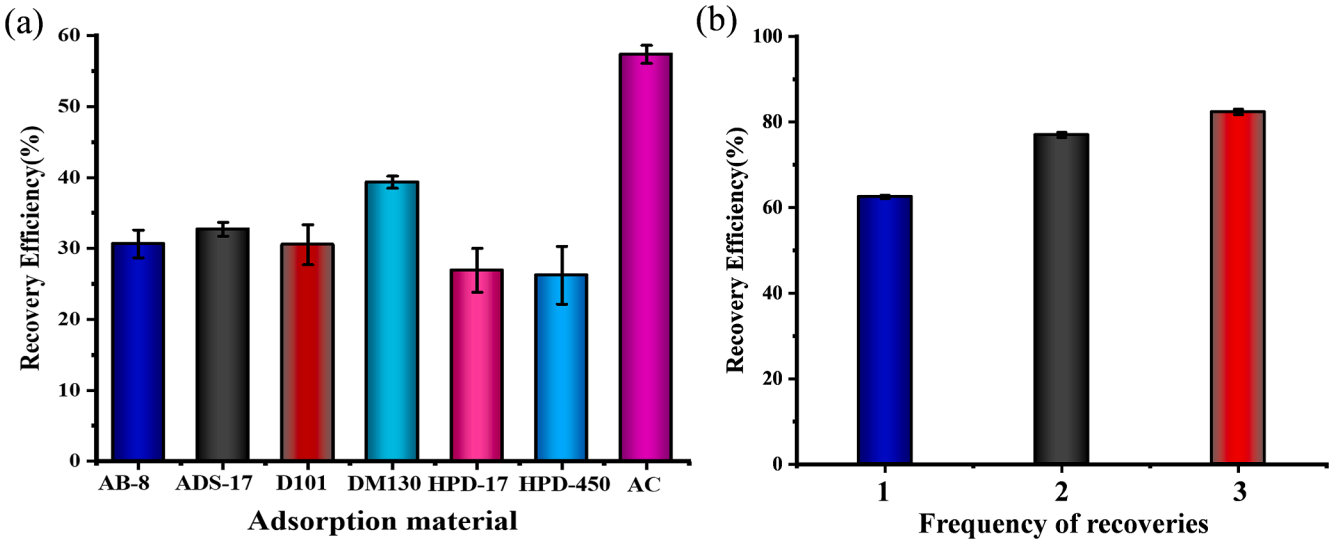


Fig. 4. The effect of different adsorbent material (a) and adsorption times of methanol soaked AC (b) on the recovery efficiency of SFN from N_{8881}Cl-EG DES extract.

3.6. Recovery of SFN from DES extract

With the advantages of cheap and simple operation, resin-adsorption methods were used to recover bioactive compounds from DES extracts per literature precedent (Cao, Yang, Cao, Wang, & Su, 2017b). Six resins (D101, DM-130, AB-8, ADS-17, HPD-17 and HPD-450) and activated carbon (AC) were adopted to recover SFN from the N_{8881}Cl-EG DES extract. The results (Fig. 4a) showed that all the resins had unsatisfactory recovery efficiency (<39.3 %), and AC showed the highest recovery efficiency (about 57.3 %).likely due to the stronger hydrophobicity of AC, which made it easier to disperse in hydrophobic DES and easily adsorb SFN. In order to achieve a higher recovery yield, various activated ACs were prepared and adopted to recover SFN (the detailed preparation methods are shown in appendix). The results (Fig. S12) showed that methanol soaking could improve the recovery efficiency of SFN from 57.3 % to 62.6 %, and three-time adsorption by methanol-soaked AC could recover more than 82.5 % of SFN (Fig. 4b). In addition, the HPLC chromatographic analysis (Fig. S13) shown that the majority of SFN in the N_{8881}Cl-EG DES extract was stripped after three-time adsorption by methanol-soaked AC (Fig. S13b and 13c). Additionally, the methanol solution (95 %, v/v) could elute SFN from AC with reduced impurities (Fig. S13b, the reduced peak intensity corresponding to impurities are found in the range of 6–11 min).

3.7. Investigation of extraction mechanism

DESs had been characterized by K-T parameters in literature, including three solvent parameters, namely H-bond donating ability (α), H-bond acceptor ability (β), and polarizability (π^*) (Hong, Shen, Xue, Sun, & Yuan, 2020). In this present study, K-T parameters of DESs and organic solvents were measured and the results (Table S15) showed that α (0.534–0.934) and β values (0.585–0.699) of DESs were obviously larger than that of traditional organic solvents (0.003–0.531 for the α value, 0.111–0.398 for the β value), which indicated these hydrophobic DESs could form stronger H-bonds with SFN than traditional organic solvents (DCM, ethyl acetate and *n*-hexane). To explore key parameters for the extraction of SFN, the correlation between the K-T parameters and the EE of SNF were conducted. The results showed that there was no obvious correlation between EE and α or π^* (Fig. 5a and c), and only β showed a positive correlation with EE of SFN (Fig. 5b), indicating higher H-bond acceptor ability was beneficial for extraction of SFN by hydrophobic DESs. Theoretically, the H-bond acceptor capacity of the solvent is connected with the H-bond donating capacity of solute. Therefore, the efficient extraction requires the synergetic effects of DESs and solute, related to both hydrogen bond and other interactions, such as electrostatic interactions, hydrophobic interactions and van der Waals forces, etc (Cao, Cao, Wang, Chen, Cao, & Su, 2020).

To explore the interaction between SFN and DESs, a theoretical analysis using DFT was conducted. After optimization of HBD, HBA, SFN and DES structures, the counterpoise corrected interaction energies of the DES + SFN system for the final optimized structures were calculated along with the HOMO (E_{HOMO}) energy, LUMO (E_{LUMO}) energy and the HOMO-LUMO energy gap (D_{EG}). The obtained results (Table S16) showed that all of the total calculated interaction energies (E_{int}) of solvents were negative (-0.173 to −0.9608 eV), indicating an exothermic interaction between the SFN and different solvents. The absolute value of E_{int} of hydrophobic DESs with SFN (0.4944 to 0.9608 eV) were obviously higher than that of NHA (0.173 eV) and ECA (0.2661 eV),

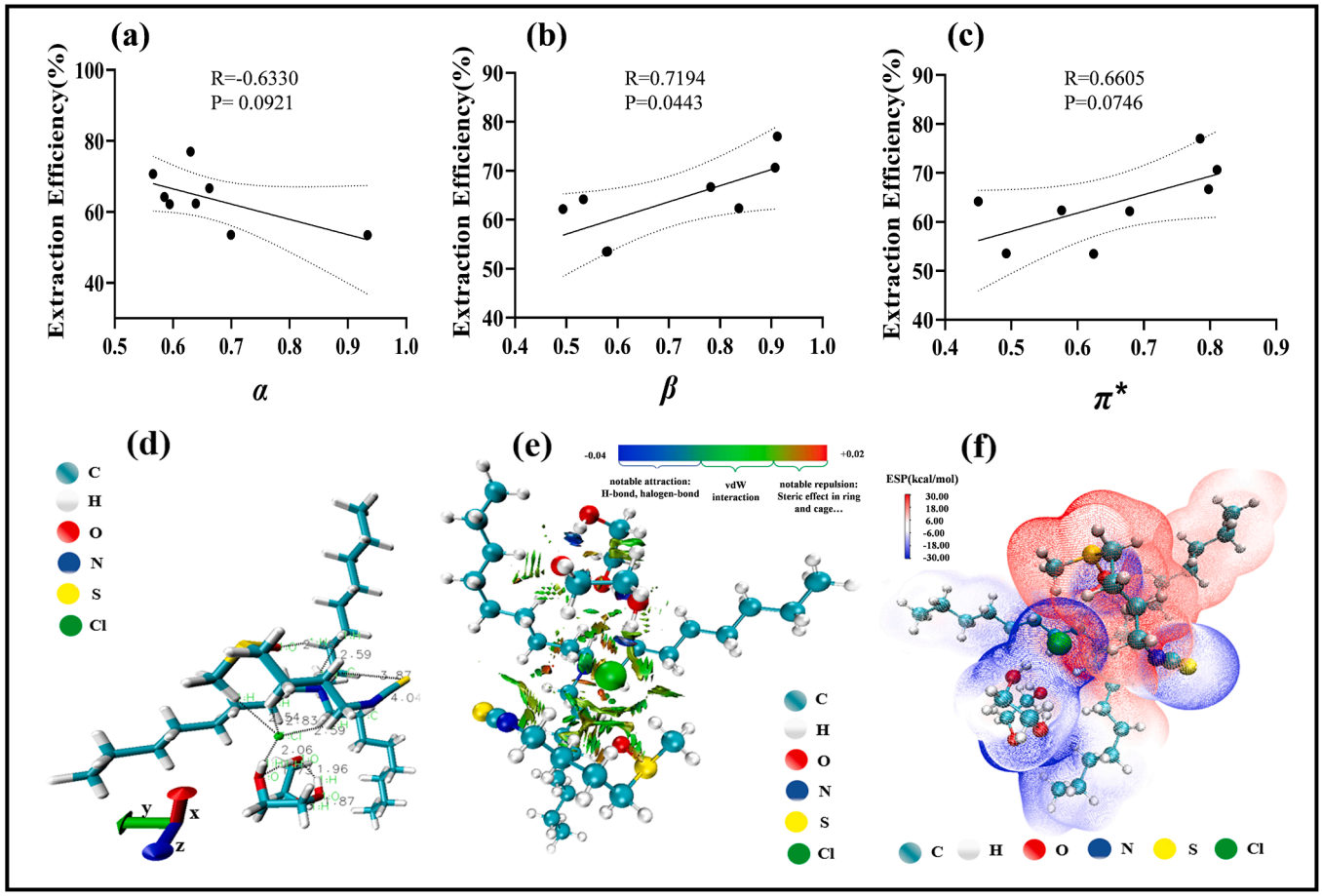


Fig. 5. The correlation between the EE of SFN and the subtraction value of α (a), β (b) and π^* (c) of hydrophobic DESs; the optimized structure (d), RDG isosurfaces (s = 0.6 a.u.) (e) and ESP mapped (0.001a.u) on electron total density (f) of N_{8881}Cl-EG DES + SFN complex calculated by DFT.

which indicated the stronger interaction between DESs and SFN (Gutiérrez, Aparicio, & Atilhan, 2019). Among the eight hydrophobic DESs, N_{8881}Cl-EG DES had the highest absolute value of E_{int} (0.9608 eV) with SFN, which was consistent with its excellent extraction performance of SNF due to its strong interaction. In addition, the positive correlation (R of 0.8793, p value of 0.0047) was observed between the EE of SFN and the E_{int} of DESs + SFN complex (Fig. S14), verifying that the interaction between DESs and SFN played a crucial role for SFN extraction.

To explore the bonding types and interaction strength between DES and SFN, reduced density gradient (RDG) and electrostatic potential field (ESP) analysis under its optimized structure were conducted with N_{8881}Cl-EG DES as a representative DESs. The obtained results (Fig. 5d) showed that non-bonded interactions exist between SFN and Cl^- of DES such as C—H···Cl (2.54 Å) and C—H···Cl^- (2.59 Å), which could be identified as the electrostatic bond due to short distance between atoms (Makoś & Boczkaj, 2019). On the basis of distance, it can be concluded that there is also a weak electrostatic bond between S═O of SFN and N_{8881}Cl with the S═O···H—C of 2.46 Å. In addition, the short distance between the SFN methylene H and N_{8881}Cl (2.59 Å) indicated the existence of a hydrophobic interaction between DES and SFN. However, there were no hydrogen bonds between EG and SFN molecule, which might be due to steric hindrance. RDG analysis results (Fig. 5e) indicated that van der Waals interactions (vdW, between C—H of SFN and Cl^- of N_{8881}Cl, S═O of SFN and C—H of N_{8881}Cl), as well as steric effects occurred in the N_{8881}Cl-EG DES + SFN complexes (Emamian, Lu, Kruse, & Emamian, 2019), and were consistent with the conformational analysis. ESP analysis (Fig. 5f) demonstrated that the large electronegative area was located around the Cl^- of N_{8881}Cl and the O atom of SFN, whereas other positive area was close to N^+ and C—H of N_{8881}Cl and C atom of SFN. The overlapped charged electron regions of SFN and DES confirmed - electrostatic interaction between DES and SFN (such as electronegative areas of O atom and S atoms in SFN with electropositive areas of C—H and N^+ in N_{8881}Cl, respectively; electropositive areas of C—H in SFN and Cl^- in N_{8881}Cl) (Makoś, Słupek, & Gębicki, 2020). According to the above results, the H-bond acceptor ability of DES and the existing vdW interaction, and electrostatic interaction between DES and SFN were crucial for the extraction of SFN from broccoli hydrolysate by hydrophobic DES.

4. Conclusion

In the present work, a novel salting-out assisted hydrophobic DES extraction method was developed and verified for its extraction performance for the weak hydrophobic SFN from broccoli. Compared to conventional organic solvents (NHA, EAC and DCM), N_{8881}Cl-EG DES-based liquid–liquid extraction exhibited excellent extraction performance for SFN, while also being considered a green solvent system.. Furthermore, the salting-out effect of acid or neutral salt significantly improved the extraction efficiency of SFN by N_{8881}Cl-EG DES. After systematic optimization by OVAT and RSM, a 97.77 % EE for SFN was achieved by N_{8881}Cl-EG DES with 1:1 of HBA/HBD, 0.25 g/mL of KH_2PO_4 and 1:1 of V_{DES}/V_{BH}. Moreover, SFN was successfully recovered from DES extracted by AC, with methanol soaking improving the recovery efficiency of SFN by AC. Following optimization, more than 82.5 % of SFN was recovered by methanol soaked AC applied in triplicate. By fitting the EE of SFN and the K-T parameter value of DESs, we found that the H-bond acceptor ability of DES was critical for extraction of SFN. DFT results indicated that the superior SFN extraction efficiency of N_{8881}Cl of DES was due to the existence of van der Waals interaction and electrostatic interactions between DES and SFN. This work systematically explored the extraction performance and its underlying mechanism of a new hydrophobic DES-based extraction method on SFN for the first time, which paves the way for the application of hydrophobic DES in the extraction and purification of other weakly hydrophobic active ingredients from plants.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgements

We thank the Hubei Provincial Department of Education (Q20201406), Hubei University of Technology (BSQD2019033) and National Natural Science Foundation of China TUBITAK (Turkey)-NSFC (China) Bilateral Cooperation Grant (82061138005) for financially supporting this project.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.foodchem.2022.134817.

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